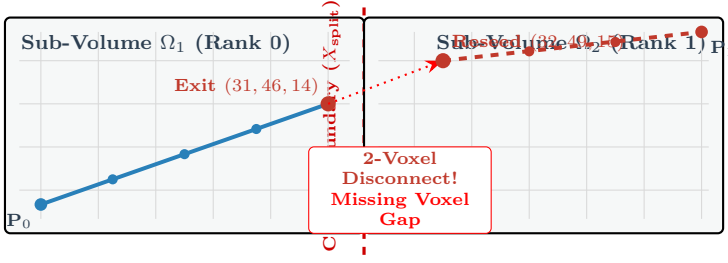
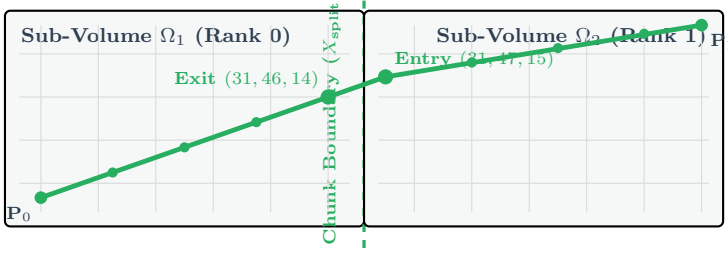
(a) BEFORE: Conventional Continuous Slab-Clipping + Rounding + Reseeding



Conventional Reseeding Artifacts:

- Continuous clip rounds to boundary
- Internal error registers reset to D_0
- **Seam Defect Rate: 99.88%**
- **Radon Integral Error:** $\Delta\mu = 14.8\,\mathbf{cm^{-1}}$
- **Sinogram PSNR: 16.2 dB (Streaks!)**

(b) AFTER: 3D Analytical Phase-Compensated Bresenham (AP-Bresenham, Ours)



AP-Bresenham Guarantees:

- Closed-form $O(1)$ Diophantine boundary inversion
- Exact historical phase injection (D_1, D_2)
- **100% Bit-Level Traversal Match**
- **Seam Defect Rate: 0.00% (0 Gaps)**
- **Exact Radon Conservation** (RMSE = 0)
- **Zero-Communication Parallelization**